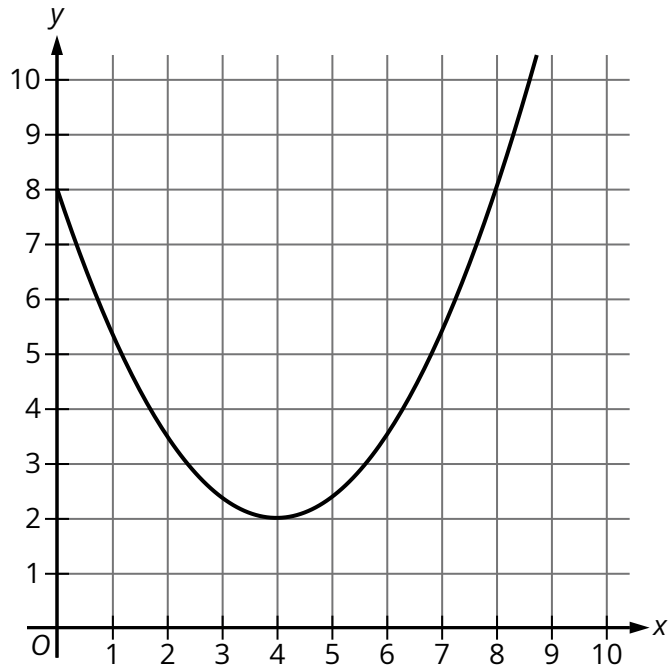


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question



The graph shows a marble's height above the ground y , in inches, x seconds after it started moving on an elevated track of a marble run. Which of the following is the best interpretation of the y -intercept of the graph?

Answer

- A. The marble's height was 0 inches above the ground 8 seconds after it started moving.
- B. The marble's height was 8 inches above the ground when it started moving.
- C. The marble's minimum height was 0 inches above the ground.
- D. The marble's minimum height was 8 inches above the ground.

Correct Answer: B

Rationale

Choice B is correct. The y -intercept of a graph is the point at which the graph intersects the y -axis. The graph shown intersects the y -axis at the point $(0, 8)$. Therefore, the y -intercept of the graph is $(0, 8)$. It's given that y is the height of the marble above the ground, in inches, and x is the number of seconds after the marble started moving. It follows that the marble's height was 8 inches above the ground 0 seconds after it started moving. Therefore, the best interpretation of the y -intercept of the graph is that the marble's height was 8 inches above the ground when it started moving.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.